

# Amazon Listing Automation

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Exchange Rate: 1 USD = ₹95

AI-Powered Catalog Enrichment Platform

## PROJECT OVERVIEW

This platform automates the end-to-end creation of Amazon-compliant product listings at catalog scale. Given a product image and SKU, the system produces a structured listing — title, five bullet points, description, HSN code, keywords — and generates up to three professional marketing images per product, all without human intervention.

## WHY GROQ API

- Only publicly-priced **vision-capable** model on Groq's infrastructure
- Llama 4 Scout 17B delivers **594 tokens/second** on LPU silicon — ~10x faster than equivalent GPU-hosted models
- One-shot architecture:** single API call returns both visual product analysis and full listing copy, halving network round-trips
- Free tier covers up to 100 vision requests/day — **zero cost for prototyping**
- 128k context window handles large constrained-fields guides from Amazon category templates
- Structured JSON output reliability at 17B scale — critical for deterministic field mapping

## WHY FAL (fal.ai) API

- FLUX.1-schnell delivers production-quality images at **\$0.003/image** — the lowest price point for this quality tier
- Serverless, zero-ops:** no GPU provisioning, auto-scales from 1 to 20,000 SKUs without configuration
- 4-step inference on schnell model — **2–5 seconds per image**, compatible with real-time batch pipelines
- Theme-aware prompts (wellness, gaming, food, baby, etc.) produce on-brand lifestyle scenes without manual art direction
- PIL fallback always available — **zero dependency failure risk** if FAL is unreachable

## MODELS & PUBLISHED RATES

| API                      | Model ID                                  | Input                         | Output / Unit                                 |
|--------------------------|-------------------------------------------|-------------------------------|-----------------------------------------------|
| Groq (Vision + One-Shot) | meta-llama/llama-4-scout-17b-16e-instruct | \$0.11 /1M tokens (₹10.42/1M) | \$0.34 /1M tokens (₹32.20/1M)                 |
| Groq (Text Fallback)     | llama-3.1-8b-instant                      | \$0.05 /1M tokens (₹4.74/1M)  | \$0.08 /1M tokens (₹7.58/1M)                  |
| FAL (Image Gen)          | fal-ai/flux/schnell                       | —                             | \$0.003/image (₹0.284/image) 4-step inference |

## API COST BREAKDOWN — PER SKU

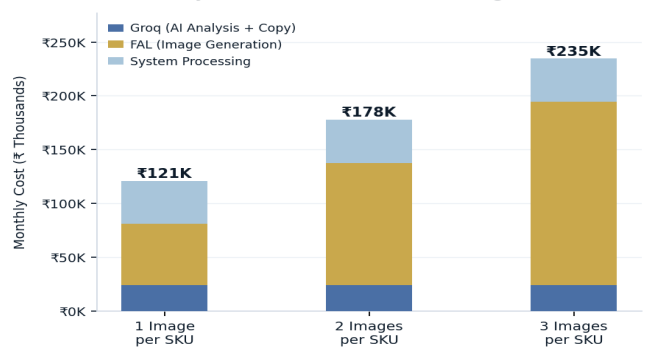
| Component      | Calculation                                | Per SKU (₹) |
|----------------|--------------------------------------------|-------------|
| Groq — Input   | 3,500 tok × \$0.11/1M × ₹94.71             | ₹0.036      |
| Groq — Output  | 2,600 tok × \$0.34/1M × ₹94.71             | ₹0.084      |
| Groq Subtotal  | Vision analysis + listing copy (1 AI call) | ₹0.120      |
| FAL — 1 Image  | 1 × \$0.003 × ₹94.71                       | ₹0.284      |
| FAL — 2 Images | 2 × \$0.003 × ₹94.71                       | ₹0.568      |
| FAL — 3 Images | 3 × \$0.003 × ₹94.71                       | ₹0.852      |
| System Charge  | Fixed processing charge/SKU                | ₹0.200      |
| TOTAL (1 img)  |                                            | ₹0.604      |
| TOTAL (2 imgs) |                                            | ₹0.888      |
| TOTAL (3 imgs) |                                            | ₹1.173      |

## MONTHLY COST SUMMARY — 2,00,000 SKUs / Month

| Cost Component            | 1 Image   | 2 Images  | 3 Images  |
|---------------------------|-----------|-----------|-----------|
| Groq (AI Analysis + Copy) | ₹24,037   | ₹24,037   | ₹24,037   |
| FAL (Image Generation)    | ₹56,826   | ₹1,13,652 | ₹1,70,478 |
| System Processing         | ₹40,000   | ₹40,000   | ₹40,000   |
| GRAND TOTAL / MONTH       | ₹1,20,863 | ₹1,77,689 | ₹2,34,515 |
| Effective Cost / SKU      | ₹0.604    | ₹0.888    | ₹1.173    |

★ Groq prompt caching (50% off static input tokens) saves ~₹1,041/month at 2,00,000 SKUs.

Monthly Cost at 2,00,000 SKUs — Image Variation



Per-SKU Cost Breakdown

